Title:

How Do Jumpscares Affect Word Search Performance

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Introduction:

The human brain, being wired to detect and respond to threats, exhibits remarkable capacity in responding to sudden, startling stimuli. Jumpscares, defined as abrupt and sudden stimuli, serves as a potent example of the human body's evolutionary response to sudden auditory and visual arousal. Scientific literature helps explain why this occurs. Specifically, according to OSU research, getting startled has been called an “amygdala hijack”, triggering the fight-or-flight response. This leads to increased cortisone, adrenaline, and mental focus. Specifically, In the words of the study's author, this heightened mental focus equips individuals to "respond more quickly to the danger they're facing." With this information in mind, this study aims to see if being jumpscared has a significant effect in enhancing subsequent cognitive tasks, such as word search performance. By investigating the duration and intensity of the alert state induced by jumpscares, we seek to understand whether this heightened cognitive arousal translates into improved attention, focus, and information-processing.

Methods:

In this experimental study, the participants were made up of family members (4/30) and students at Wesleyan (26/30). The covariates among the participants were Biological sex (M or F with 25 M and 5 F) and Underclassmen (Yes or No with 11 Yes and 19 No). The data was collected both at home, where I instructed my family members to come one at a time and at the Science Library in Exley, where I used a study room to test participants one by one (the room where the study took place was downstairs and participants were waiting for their turn upstairs). The data was collected from 11/26 to 11/29. Using an online software, the 30 participants were equally distributed into 3 categorical conditions: no jumpscare (NJ), jumpscare with audio (J) and jumpscare without audio (JNA). The chosen jumpscare for the experiment was the "Scary Maze Game," an online game that became an internet sensation during the 2010s, known for its powerful and unexpected jumpscare.

The "Scary Maze Game" operates with the following procedures. It first starts out with a prompt stating to the player "can you beat all 4 levels." Once the player clicks the black play button, the user is directed to level 1. In the game, there is a maze compiled of 3 colors: black, blue and red. The player has to get to the red square at the end of the maze while staying within the blue and not touching the black. Level 1 is easy, level 2 is a bit harder and level 3 is the most challenging. Level 3, however, has an interesting twist. Once the player reaches the last corner, the maze narrows causing the player to slow down. Once the cursor arrives at a certain point, the player is jumpscared both visually and auditorily by a screaming devil child. Despite the prompt stating there are 4 levels, the game, being the troll it is, only contains the aforementioned 3 levels. From the three experimental conditions, the control group (NJ) did not play the game, the J group played normally and the JNA group played the game with the audio muted.

To assess cognitive performance in the aftermath of the jumpscare, a word search activity was employed as a measure. This 15x15 letter word search, titled the Ocean Animals Word Search, contains 15 ocean-themed words. A physical copy of the word search was printed out for each participant and placed face-down next to them. The experimental procedure instructed the participant to flip the piece of paper over either immediately (for NJ) or after experiencing the jumpscare (JNA and J). This variation in experimental procedures aimed to produce different levels in cognitive engagement (and thus performance) between the three experimental groups.

Upon flipping the piece of paper over, each participant was given 60 seconds to find (or solve) as many words as possible out of the 15 presented in the word bank. This quantitative measure served as the study's cognitive performance metric, assisting in the evaluation of the impact of jumpscares on participants and their ability to concentrate after being subjected to the jumpscare, measured by their ability to find the hidden words in the puzzle (solve the words) within the given time constraint.

By combining a word search with a jumpscare, the study aimed to examine the effect of completing a cognitive task in a heightened state of visual and auditory stimulation. This approach provides insights if it coincides with the known literature on jumpscares and neurological impact.

Results:

The results of the experiment can be best visualized using a boxplot. Across the entire study, the grand mean for the number of words solved was 3.367 with a standard deviation of 1.5 words. When comparing among the three groups, the NJ group had a mean score of 2.6 words with a standard deviation of 1.43 words, the J group had a mean score of 3.6 words with a standard deviation of 1.71 words and the JNA group had a mean score of 3.9 words with a standard deviation of 1.1 words. The highest and lowest score for each group was 7 and 1 for the J group, 6 and 2 for the JNA and 0 for the NJ group.

In order to test for significance between groups, an ANOVA significance test was conducted with the covariates Sex and Underclassmen. The results from the ANOVA test are as follows: Since the p value is 0.139 (>0.05), ($F(2,25) = 2.138$), there are no statistically significant differences in the mean number of words solved between the different assignment conditions. For the sex covariate, since the p value is 0.436 (>0.05), ($F(1,25) = 0.626$), there is no statistically significant difference in the number of words solved based on sex after controlling for the other variables. For the underclassmen covariate, since the p value is 0.792 (>0.05), ($F(1,25) = 0.071$), there are no statistically significant differences in the number of words solved based on whether participants are underclassmen or not, after controlling for the other variables.

Following the ANOVA test, a *post hoc* test was utilized to determine if results varied significantly among the three groups. The results for the *post hoc* test are as follows: since the value 0.664 is greater than 0.05, suggesting that there is not enough evidence to reject the null hypothesis for the comparison between the jump scare group and the jumpscare no audio group. For the no jumpscare group and jumpscare group, since the value 0.131 is greater than 0.05, suggesting that there is not enough evidence to reject the null hypothesis for the comparison between the no jump scare group and the jump scare group. For the no jumpscare group and jumpscare no audio group, since the value 0.053 is greater than 0.05, suggesting that there is not enough evidence to reject the null hypothesis for the comparison between the No jump scare group and the jump scare no audio group.

Discussion:

Although the mean word search scores exhibited a trend to be higher among the groups with jumpscares (as supported by the research), the results failed to achieve statistical significance. Several factors could contribute to the lack of significance, one of which being uneven distribution of covariates. In the study, the sex covariate had 5 female and 25 male participants and the underclassmen covariate had 11 underclassmen and 19 non-underclassmen. The imbalances in these covariate groups could affect the reliability and significance of the ANOVA test results.

Surprisingly, the JNA group outperformed the other two groups in mean score and had 1) a smaller standard deviation and 2) the lowest p value in the post-hoc test. The JNA group did seem less scared than the J group, so it is surprising that they achieved such scores given the literature.

Additionally, the study did contain a notable confounding variable, being prior exposure to the "Scary Maze Game" in the past (2 participants out of 30). The presence of a confounder reduces the effectiveness of the jumpscare stimuli, thereby affecting the significance of the data.

Despite the non-significant results of this experiment, with a better experimental design, I believe that the results would produce more meaningful outcomes. One key modification would involve increasing the sample size. A larger and more diverse participant pool would likely enhance the likelihood of achieving statistically significant p values. Another change that I would make would be adjusting either the word search puzzle itself or the time period for solving the puzzle. While 60 seconds is an appropriate amount of time to allow participants to solve the word search while still being under the influence of the jump scare, since the highest score among the 30 participants was 7 words, the test may be improved by increasing the testing time. Also, most participants reported that they did not see the word bank at first. Additional adjustments such as simplifying the words, enlarging the font, and extending the time allotted for completion could possibly improve the experimental design

To conclude, while the current study did not achieve statistical significance, a careful reconsideration could yield more insightful results. Notably, variation in reaction to the jumpscare – including screaming, being startled and no reaction at all – suggests the potential for the investigation of emotional responses in a larger and more diverse sample. As such, future research could reveal greater insights regarding emotional response and jumpscares.

Appendix:

OSU study: Ohio State University Health Services. "Psychology of Scary Movies." OSU Health, https://health.osu.edu/health/mental-health/psychology-of-scary-movies.

Random Word Generator: https://www.dcode.fr/random-selection

Scary Maze Game: https://www.silvergames.com/en/scary-maze

```
#Final Proj

library(readxl)

Jumpscare <- read_excel("Desktop/Data Analysis/Jumpscare.xlsx")

View(Jumpscare) # if the data set does not load, download link
```

https://docs.google.com/spreadsheets/d/1Z6dDT5iAmNjo8gQ6kaIWleOChC1ntYWy0yyl-uGCpcM/edit?usp=sharing

```
library(ggplot2)

library(dplyr)



Jumpscare %>%

  mutate(Underclassmen=ifelse(Underclassmen=="no",0,1))-> Jumpscare


Jumpscare %>%

  mutate(Sex=ifelse(Sex=="F",0,1))-> Jumpscare


Jumpscare <- Jumpscare %>%

  mutate(Assignment_new = as.numeric(factor(Assignment, levels = c("nj", "j", "jna"))))


tapply(Jumpscare$Score, Jumpscare$Assignment, FUN=mean)


mean(Jumpscare$Score)


sd(Jumpscare$Score)


tapply(Jumpscare$Score, Jumpscare$Assignment, FUN=sd)
```

```
model=aov(Score~Assignment, data=Jumpscare)

summary(model)


#Since the p value is 0.125 (>0.05), (F(2,27) = 2.246),

#we fail to reject the null hypothesis.

#there are no significant differences in the mean number of words

#solved between the different Assignment conditions.


model2=aov(Score~Assignment+Sex+Underclassmen, data=Jumpscare)

summary(model2)


#Since the p value is 0.139 (>0.05), (F(2,25) = 2.138),

#we fail to reject the null hypothesis.

#there are no significant differences in the mean number of words

#solved between the different Assignment conditions.


#Since the p value is 0.436 (>0.05), (F(1,25) = 0.626),

#his suggests that there is no statistically significant difference

#in the number of words solved based on sex after controlling for

#the other variables.


#Since the p value is 0.792 (>0.05), (F(1,25) = 0.071),
```

```
#This suggests that there is no statistically significant difference in the number
#of words solved based on whether participants are underclassmen or not, after
#controlling for the other variables.

pairwise.t.test(Jumpscare$Score, Jumpscare$Assignment, p.adj="none")

#the value 0.664 is greater than 0.05, suggesting that there is not enough evidence
#to reject the null hypothesis for the comparison between the jump scare group
#and the jumpscare no audio group.

#the value 0.131 is greater than 0.05, suggesting that there is not enough evidence
#to reject the null hypothesis for the comparison between the No jump scare group
#and the jump scare group.

#the value 0.053 is greater than 0.05, suggesting that there is not enough evidence
#to reject the null hypothesis for the comparison between the No jump scare group
#and the jump scare no audio group.

ggplot(data=Jumpscare)+
  geom_boxplot(aes(x=Assignment, y=Score))+
  ylab("Words Found")+
  xlab("Group Assignment")+
  ggtitle("# of Words Found Based on Assignment")
```

Word Search:

Ocean Animal Word Search

S	L	S	E	Y	S	S	S	X	V	Z	E	E	C	Q
M	E	B	U	Q	N	T	H	J	H	S	N	R	Z	R
C	E	E	U	P	I	S	A	Y	A	E	O	N	J	S
S	S	I	R	N	O	B	R	L	I	S	M	N	R	T
J	D	E	G	O	Z	T	K	J	C	Y	E	L	I	A
G	E	R	M	A	L	C	C	Y	T	B	N	B	D	R
M	A	L	N	I	H	P	L	O	D	X	A	E	R	F
Y	V	S	L	O	B	S	T	E	R	T	E	R	T	I
X	U	D	B	Y	Y	C	F	E	Y	W	G	M	C	S
P	W	F	Z	Y	F	J	N	L	A	O	U	U	P	H
U	N	H	Q	W	L	I	B	E	R	H	B	B	M	L
Y	L	J	A	D	E	F	S	Z	Y	F	Y	I	G	F
J	F	M	G	L	C	L	M	H	F	V	Z	F	K	M
P	T	W	B	J	E	N	T	D	B	Q	X	H	A	T
S	W	O	R	D	F	I	S	H	U	D	C	W	Q	C

ANEMONE	**CLAM**	**CRAB**
DOLPHIN	**EEL**	**JELLYFISH**
LOBSTER	**OCTOPUS**	**SEAWEED**
SHARK	**SQUID**	**STARFISH**
STINGRAY	**SWORDFISH**	**WHALE**